

Iyed MEDIOUNI

AI ENGINEER · LLM SYSTEMS & APPLIED MACHINE LEARNING

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Portfolio iyed122.github.io



PROFILE

AI Engineer building production-grade LLM systems end to end — hybrid Graph-RAG retrieval over Neo4j, a LangGraph multi-agent orchestrator with live API routing, and a self-improving evaluation loop that gates QLoRA/DPO fine-tuning behind measured quality gains. Equally comfortable in the modelling and the plumbing: on-premise inference, FastAPI services, retrieval design, and the operator tooling that makes an assistant usable day to day. Shipped work spans multi-agent systems, computer-vision pipelines and applied security research.

EXPERIENCE

02/2026 – 06/2026

Lac 1, Tunis

AI Engineer Intern — TELNET Holding

- Project knowledge sat fragmented across Jira, Confluence and GitLab — delivered a production, on-premise assistant on **LangGraph** and **Neo4j Graph-RAG** that answers questions across all three from one natural-language interface.
- Early responses were too slow to feel interactive: built a hybrid query-routing layer pairing deterministic classification with parallel API dispatch, cutting mean latency **2.4x** (7.3s → 3.0s).
- Fine-tuning was running blind with no measure of answer quality — designed **Hammer**, an evaluation framework (RAGAS + deterministic validation) that grades every answer on a four-tier scale crossed with a failure taxonomy, so no **QLoRA/DPO** run ships without a proven gain.
- Traced a silent scoring defect that mislabelled correct answers as failures and was poisoning the training set; the fix lifted training-data quality **~10 points** and model faithfulness **27%**.
- Took the system from prototype to something a team can run: **FastAPI** training services, MongoDB persistence, local inference via **Ollama**, and a **React/Vite** console for training mode, DPO curation and QLoRA candidate review.

PROJECTS

07/2025 – 10/2025

Autonomous Vehicle Perception & Trajectory Prediction

REPO github.com/iyed122/autonomous-vehicule-VLM

- Camera detection alone can't anticipate where an object is heading — built an end-to-end pipeline on **AWS SageMaker** pairing a fine-tuned **YOLOv8** detector with a custom bird's-eye-view model for LiDAR trajectory prediction, trained in PyTorch to **ADE 0.8** / **FDE 1.2**.
- Made the output legible to non-specialists through a **Streamlit** dashboard surfacing live detection, trajectories and a CLEAR / CAUTION / BRAKE collision-risk signal across 6-camera and LiDAR views.
- Integrated a local vision-language model (**Qwen2.5-VL**) that narrates driving scenes over the full sensor suite, turning raw detections into structured hazard analysis and automated commentary.

09/2025 – 11/2025

CrewAI Multi-Agent Content Automation

REPO github.com/iyed122/automated-twitter-content

- Generic LLM posts drift from what is actually trending — built a **CrewAI** + **LLaMA-3.1-8B** pipeline where specialised agents research live web, search and trend data before any drafting begins.
- Closed the loop to publication via the X API with secure OAuth, plus a grounding step tying every generated post back to its retrieved sources.

EDUCATION

2023 – 2026

Engineering Degree — Data Science & Artificial Intelligence

TEK-UP University

2020 – 2023

Bachelor in Computer Science

ISAMM, Manouba

TECHNICAL SKILLS

AI & ML	LangGraph · LangChain · CrewAI · Graph-RAG · RAGAS · QLoRA / DPO fine-tuning · Ollama · PyTorch · TensorFlow · Scikit-learn · OpenCV
LANGUAGES	Python · JavaScript · SQL · PL/SQL · R · MATLAB
DATA & BACKEND	Neo4j · Qdrant · MongoDB · PostgreSQL · MySQL · Cassandra · FastAPI · Pandas · NumPy
CLOUD & MLOPS	AWS SageMaker · EC2 · Lambda · S3 · DynamoDB · Redshift · RDS · Docker · MLflow
FRONTEND & TOOLING	React · Vite · Streamlit · Git / GitLab · LaTeX
SECURITY	Penetration-testing fundamentals · static analysis of client-side JavaScript · secure API & OAuth integration

LANGUAGES & CERTIFICATIONS

ENGLISH	IELTS Academic, British Council — B2 (6.5 / 9)
FRENCH	Professional working proficiency — academic and technical writing